Peiyu (Georgia) Li

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EDUCATION

University of Notre Dame, Notre Dame, USA *Spring 2026 - Present*

Ph.D. Computer Science and Engineering

Advisor: ***Prof. Nitesh V. Chawla*** (ACM/IEEE/AAAI/AAAS Fellow)

University of Notre Dame, Notre Dame, USA *Aug. 2024 - May. 2025*

MSc Computer Science and Engineering GPA: 3.93/4.0

Sun Yat-sen University, Guangzhou, China *Aug. 2019 - Jun. 2023*

BSc Philosophy, (Logic & Reasoning Focus) GPA: 3.9/4.0 (92/100) Ranking: Top 4

SKILLS

Languages: Python, C++, Java, SQL, R, JavaScript

ML & Systems: PyTorch, RAG, LangChain, LLM fine-tuning, Multimodal Models, Agents, Deepspeed

Tools & Platforms: AWS (EC2/S3/Lambda), Docker, Linux, Git, REST APIs

SELECTED RESEARCH & PROJECTS

Adaptive Testing for LLM Evaluation | *Machine Learning Evaluation, Data Sampling, R* **|** Github

- **Developed an adaptive LLM evaluation** achieving **90% reduction in test items** with **0.154 MAE**, delivering **8.7× faster evaluation** with equivalent leaderboard fidelity.
- **Analyzed 4,000+ models across 6 benchmarks**, identifying 23–31% ranking instability due to low-discrimination items.

Executable Benchmark and Generation Framework | *LangChain, Multi-Agent Systems* | Github

- **Developed CrochetBench**, the first *executable benchmark* for procedural reasoning in multimodal (vision–text–DSL) crafts, unifying symbolic, textual, and visual modalities.
- Built a **multi-agent generation system** (Designer → Critic → Publisher) for automatic crochet pattern creation, improving executable validity from **48% → 83%** on DSL-based evaluation.

On-Device AR LLM System | *Embedded AI, Quantization, Edge Computing* | Video

- **Built offline real-time STT AR glasses system** integrating ESP32-S3 (audio capture), Raspberry Pi 4, and Brilliant Labs Frame (display), achieving **< 5s latency per 6s speech** segment **with < 210MB memory usage**.
- **Reduced inference latency by 42%** through Whisper.cpp quantization and thread optimization, achieving real-time, privacy-preserving on-device STT.

TeachingCoach — Pedagogical LLM Systems | *Fine-tuning, Model Deployment, Deepspeed* | Video

- **Built TeachingCoach**, a pedagogical LLM system **developed for active use by Notre Dame instructors** in collaboration with **NDLearning (Teaching Innovation Office)**.
- Fine-tuned **LLaMA-13B** with structured pedagogical data, surpassing **GPT-4o** in instructional quality.
- **Deployed on AWS** and integrated into Notre Dame's teaching toolkit, enabling **real-time classroom support and instructor-facing guidance** validated through an **HCI user study**.

Multimodal Recipe & Food Image Generation | *Multimodal, PyTorch, Big Data* | Github | CIKM 2024

- Developed a **multimodal** AI framework on **300K+** recipe–image pairs**,** integrating text-to-text, text-to-image, and image-to-text tasks, enabling real-time **interactive multimodal dialogue** via a Gradio demo.
- **Combined OPT-6.7B** and **Stable Diffusion v1.5** with a lightweight transformer using custom `[img]` tokens for efficient **cross-modal alignment** and **semantic consistency** between recipes and images.

WORK EXPERIENCE

Research Associate| DIAL Lab **|** *University of Notre Dame* *Oct. 2023 - Present*

Led end-to-end research pipelines spanning dataset curation, model training, evaluation, and deployment.

Teaching Assistant: Python in Data Science| *University of Notre Dame* *Aug. 2024 - May.2025*

Taught applied ML tools (Pandas, NumPy, visualization) to 100+ students